Problem Solving Using Python (CSE1012)

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Agenda / Contents

Nested Loop

- It is a loop inside another loop
- Most commonly used with “for loops” because it is easier to control
- The loops should be properly indented to identify which statements are contained within each loop

Nested Loop Example

Program to generate a pattern as follows:

```
* * *
* * *
* * *
* * *
```

Nested Loop Example

```
# Program to generate a pattern as follows:
for i in range(4):
    for j in range(3):
        print("*",end=' ')
    print()
```

Nested Loop Example

Prog to generate a pattern of numbers

1

1 2

1 2 3

1 2 3 4

Nested Loop Example

```
# Prog to generate a pattern of numbers
for i in range(1,4+1):
	for j in range(1,i+1):
		print(j,end=" ")
	print()
```

Nested Loop Example

```
# Prog to generate a pattern of numbers
n=int(input("Enter the number of rows to be printed: "))
for i in range(1,n+1):
    for j in range(1,i+1):
        print(j,end=" ")
    print()
```

Nested Loop Example

Prog to generate a pattern of numbers

1

2 2

3 3 3

4 4 4 4

Nested Loop Example

```
# Prog to generate a pattern of numbers
for i in range(1,4+1):
    for j in range(1,i+1):
        print(i,end=" ")
    print()
```

HW: Print the following pattern

```
        1
      1 2 1
    1 2 3 2 1
  1 2 3 4 3 2 1
1 2 3 4 5 4 3 2 1
```